

## Solution Requirements

|                      |               |
|----------------------|---------------|
| <b>Date</b>          | 6 June 2026   |
| <b>Team ID</b>       | XXXXXX XXXXXX |
| <b>Project Name</b>  | Rising Water  |
| <b>Maximum Marks</b> | 4 Marks       |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                      | Priority (High/Medium/Low) |
|------|----------------------|------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | <i>Users can securely access the web application (optional login).</i>       | <i>Medium</i>              |
| 2    | Authorization levels | <i>Normal users can predict floods; administrators can update the model.</i> | <i>Medium</i>              |
| 3    | External interfaces  | <i>Supports weather datasets and future integration with Weather APIs.</i>   | <i>Low</i>                 |

|   |                                   |                                                                                                 |               |
|---|-----------------------------------|-------------------------------------------------------------------------------------------------|---------------|
|   |                                   |                                                                                                 |               |
| 4 | Transactions processing           | <i>Accepts weather parameters, preprocesses data, predicts flood risk, and returns results.</i> | <i>High</i>   |
| 5 | Reporting                         | <i>Displays flood prediction, probability score, and input summary.</i>                         | <i>High</i>   |
| 6 | Business rules, etc.              | <i>Prediction is generated only when all required values are entered.</i>                       | <i>High</i>   |
| 7 | Compliance to laws or regulations | <i>Protect user privacy and securely handle submitted data.</i>                                 | <i>Medium</i> |
| 8 | <i>Other</i>                      | <i>Support deployment on Flask/Streamlit with a responsive interface.</i>                       | <i>Medium</i> |

**Step 2: Non-Functional Requirements (NFR)**

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|      |              |                         |                                     |

|          |                            |                                                                |                                |
|----------|----------------------------|----------------------------------------------------------------|--------------------------------|
| <b>1</b> | Performance & Speed        | <i>Prediction should be generated quickly.</i>                 | <i>&lt; 2 seconds</i>          |
| <b>2</b> | Scalability                | <i>Support increasing numbers of users.</i>                    | <i>1000+ users</i>             |
| <b>3</b> | Security & Data Privacy    | <i>Protect user data and validate inputs.</i>                  | <i>Secure input validation</i> |
| <b>4</b> | Reliability & Availability | <i>System should remain available.</i>                         | <i>99% uptime</i>              |
| <b>5</b> | Usability & Accessibility  | <i>Simple and responsive interface for desktop and mobile.</i> | <i>User-friendly UI</i>        |
| <b>6</b> | Other                      | <i>Easy to update models and retrain with new datasets.</i>    | <i>Modular architecture</i>    |